

3(a)	change of displacement/time (taken)	B1
3(b)(i)	constant velocity, so resultant force is zero	M1
	(so car is) in (dynamic) equilibrium	A1
3(b)(ii)	$F_D = 0.40 \text{ (kN) or } 0.40 \times 10^3 \text{ (N)}$	C1
	component of weight $= 2.0 \times 10^3 - 0.40 \times 10^3$ $= 1.6 \times 10^3 \text{ N}$	A1
3(b)(iii)	$P = Fv$	C1
	$= 2.0 \times 10^3 \times 9.0 = 1.8 \times 10^4 \text{ W}$	A1
3(b)(iv)	(driving) force $= 1.8 \times 10^4 / 15 (= 1.2 \times 10^3)$	C1
	$F_D = 0.66 \text{ (kN) or } 0.66 \times 10^3 \text{ (N)}$	C1
	acceleration $= (1.2 \times 10^3 - 0.66 \times 10^3) / 850$ $= 0.64 \text{ (0.635) m s}^{-2}$	A1